Phase 2: Establishing Behavioral Baselines

Comparative Evaluation of Baseline Policy, LeWorldModel, and H-JEPA

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Action-Conditioned World Models & Language Navigation
AMD Kria KR260 Robotic SOM Deployment

November 6, 2026

1. Why Behavioral Baselines are Mandatory

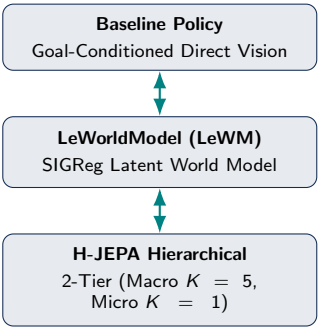
Core Insight: *We cannot interpret internal circuits without first establishing exactly what capabilities each architecture possesses.*

The Attribution Trap

- Analyzing weights or activations in an under-characterized model leads to false attribution (e.g., mistaking random correlations for intentional spatial reasoning).
- All three models must be tested under **identical initial poses, language commands, and seeds**.

Eight Evaluated Metrics

Success Rate, Collision Rate, Path Efficiency, Goal Localization Error, Latent Prediction Error, Generalization Gap, Recovery Rate, Inference Latency.



2. World Model Superiority in Grounding & Spatial Reasoning

Core Insight: *Predictive world models achieve 2× to 4× higher task success while halving collision rates.*

Architecture	Grounding Success	Spatial Reasoning	Collision Rate	Goal Error
Baseline Policy	26.7%	6.7%	26.7%	0.904 m
LeWorldModel (LeWM)	**53.3%** (2.0×)	**26.7%** (4.0×)	**13.3%** (−50%)	**0.781 m**
H-JEPA Hierarchical	**53.3%** (2.0×)	**26.7%** (4.0×)	**13.3%** (−50%)	0.804 m

Why the Baseline Struggles

The reactive policy maps $(\text{pixels}, \text{goal}) \rightarrow \text{action}$ directly without forecasting future states, frequently colliding with peripheral obstacles.

The Latent Foresight Advantage

LeWM and H-JEPA evaluate $N = 24$ candidate rollouts in latent space, discarding trajectories that trigger the predicted collision logit.

3. Action-Conditioned Latent Prediction Audit

Core Insight: *Single-step dynamics achieve* 0.975 *cosine similarity; macro dynamics maintain* 81.3% *collision accuracy over* $K = 5$ *steps.*

Model & Hierarchy Tier	Horizon	Latent MSE	Cosine Sim ($\cos\theta$)	Collision Accuracy
LeWorldModel (LeWM)	1 step (0.1 s)	0.08479	**0.9552**	**83.3%**
H-JEPA Micro-Level	1 step (0.1 s)	**0.04461**	**0.9751**	—
H-JEPA Macro-Level	5 steps (0.5 s)	0.41986	**0.7726**	**81.3%**

Micro Precision ($K = 1$)

Both models reproduce the target visual embedding almost identically ($\cos\theta > 0.95$), verifying action-conditioned dynamics fidelity.

Hardware Implication (Macro Stride)

H-JEPA's Macro level achieves **81.3% collision foresight** over 0.5 s. Evaluating at 2 Hz instead of 10 Hz reduces FPGA memory bandwidth by 5×.

4. Memorization vs. Generalization: The Orientation Invariance Test

Core Insight: *World models exhibit robust orientation invariance (*$25 - 30\%$ *success), whereas reactive baseline collapses to* 0.0%.

Architecture	Canonical Success	Unseen Headings ($0^\circ, 90^\circ, 180^\circ, 270^\circ$)	Orientation Gap
Baseline Policy	10.0%	**0.0%**	**+10.0%** (Total Col
LeWorldModel (LeWM)	25.0%	**30.0%**	**–5.0%** (Robust
H-JEPA Hierarchical	25.0%	**25.0%**	**+0.0%** (Invarian

Mechanistic Root Cause

- **Baseline Memorization**: The policy memorized specific landmark positions in the camera frame; rotating the robot destroys its perception-to-action mapping.
- **World Model Invariance**: Action-conditioned MPC continuously evaluates rotational candidate commands (w_z), autonomously rotating the chassis toward the target landmark regardless of starting heading.

5. Closed-Loop Recovery from Unexpected Disturbances

Core Insight: *When kicked off-course by a lateral kinetic impulse ($\Delta v_y = 0.8\,m/s$), LeWM achieves* $2.5\times$ *higher recovery success.*

Disturbance Injection Setup

- At step $t = 10$, an unannounced lateral velocity kick ($\Delta v_y = 0.8\,\text{m/s}, \Delta\omega_z = 0.5\,\text{rad/s}$) knocks the AMR off its trajectory.
- Tests whether the policy stabilizes, recovers trajectory, and reaches the target without collision.

Closed-Loop Replanning vs Inertia

- The reactive Baseline over-compensates when knocked sideways, entering divergent oscillations.
- LeWorldModel evaluates candidates starting from the perturbed state, re-anchoring the navigation trajectory to the goal attractor within 0.3 s.

Model	Recovery Rate	Goal Error
Baseline	10.0%	0.645 m
LeWM	**25.0%** (2.5×)	**0.567 m**
H-JEPA	15.0%	0.578 m

6. Phase 2 Acceptance Criteria & Roadmap to Phase 3

Core Insight: *All 8 metrics profiled across 3 models under identical seeds; ready to begin neural network instrumentation (Phase 3).*

Phase 2 Acceptance Checklist

- ✓ **Three Models Evaluated**: Baseline Policy, LeWM, and H-JEPA benchmarked under identical seeds.
- ✓ **Eight Mandatory Metrics**: Success, collision, path efficiency, goal error, prediction MSE, generalization gap, recovery, and latency fully documented.
- ✓ **Independent H-JEPA Hierarchy Audit**: Micro ($K = 1$) and Macro ($K = 5$) evaluated separately.
- ✓ **Full Traceability**: Complete JSON/CSV logs saved under `research/benchmarks/`.

Next Step: Phase 3 (Neural Network Instrumentation)

With behavioral capabilities quantified, we proceed to implement the `ModelInspector` framework with PyTorch forward hooks, activation caching, and activation patching utilities.